

WASP-43b

R-1354

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_220416 · created 2026-07-31T22:56:22+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459685.7182 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.057
Transit depth (Rp/Rs)^2	2.8e-05 +/- 0.0061 [%]
Orbital Inclination (inc)	80.63 +/- 2.9
Airmass coefficient 1 (a1)	0.142 +/- 0.099
Airmass coefficient 2 (a2)	1.38 +/- 0.74
Scatter in the residuals of the lightcurve fit is	78.9 %
Best Comparison Star	None
Optimal Aperture	1.71
Optimal Annulus	7.06
Transit Duration (day)	0.033 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.057 vs 0.1594 (deviation 2.78σ)
Duration fit vs published	0.033 d vs 0.0512 d (deviation 1.07σ)
Mid-time O–C	-4.3 min
Depth SNR	0.0
Residual scatter	78.9 %

Light curve

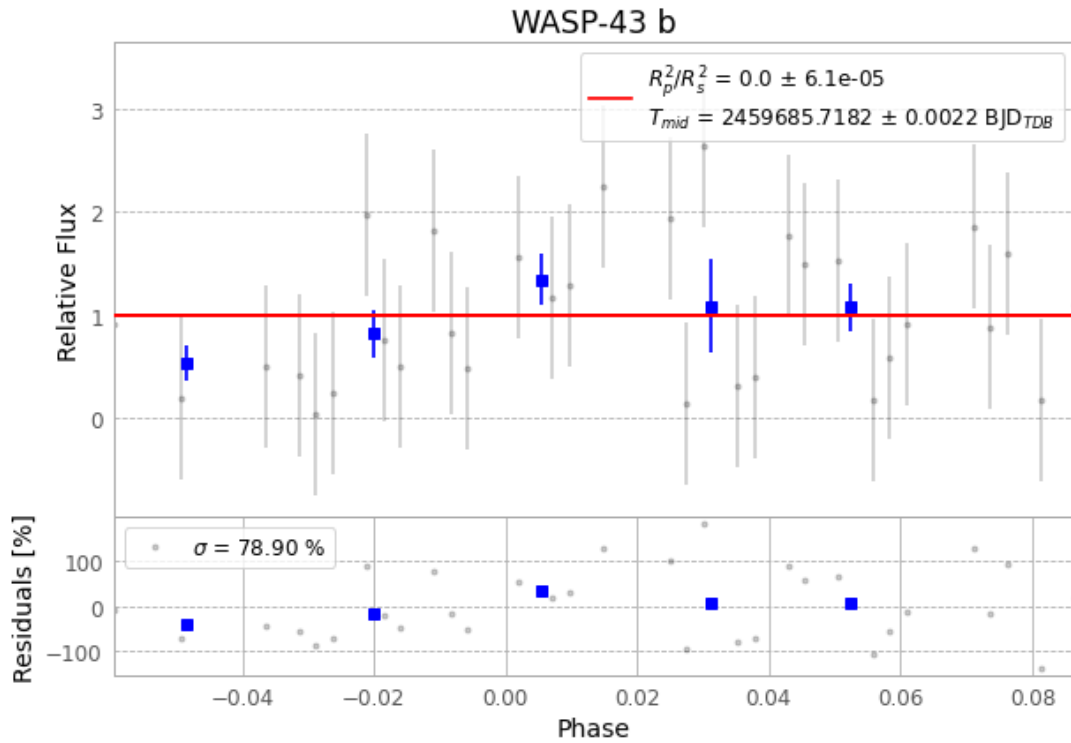


Figure 1 — FinalLightCurve_WASP-43 b_2022-04-15.png (SHA-256 ddca8af5c1c9f0fc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [550, 326], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 61e2ff7d8b573f3a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

60 FITS frames (39 MB), WASP-43220416034815.FITS → WASP-43220416064814.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-220416.log (b598be0b3a0c...), FinalLightCurve_WASP-43 b_2022-04-15.png (ddca8af5c1c9...), FinalLightCurve_WASP-43 b_2022-04-15.csv (45e56d9917a7...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 22:56Z.